AITS
ALL INDIA TEST SERIES
SELECTION HOGA YAHIN SE

Dropper JEE Main

AJPM7/06 **Test- 06**

DURATION : 180 Minutes **DATE : 19/01/2025** **M.MARKS : 300**

Topics Covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 75 questions. The maximum marks are **300**.
4. There are three Sections in the question paper, Section I, II & III consisting of Section-I **(Physics)**, Section-II **(Chemistry)**, Section-III **(Mathematics)** and having **25 questions** in each part in which first **20** questions are of Objective Type and Last **5 questions** are integers type and all **25 questions are compulsory**.
5. There is only one correct response among 4 alternate choices provided for each objective type question.
6. Each correct answer will give **4** marks while **1** Mark will be deducted for a wrong response.
7. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
8. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.
9. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ **Invigilator's Signature** _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question.***

❑❑❑

SECTION-I (PHYSICS)

Single Correct Type Questions

1. A particle is projected from origin with velocity v_0 along positive x-axis. A damping force is acting on the particle which is proportional to the cube of the distance from the origin i.e. $ma = -\alpha x^3$. The distance at which the particle stops

(1) $\left(\frac{2mv_0^2}{\alpha}\right)^{\frac{1}{4}}$ (2) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{4}}$

(3) $\left(\frac{2mv_0^2}{\alpha}\right)^{\frac{1}{2}}$ (4) $\left(\frac{3mv_0^2}{2\alpha}\right)^{\frac{1}{2}}$

2. Potential energy of a system of two atoms as a function of r is given by $U = \frac{A}{r^8} - \frac{B}{r^4}$, where r is the interatomic distance, A and B are positive constant. The equilibrium distance between the two atoms will be

(1) $\left(\frac{A}{B}\right)^{\frac{1}{4}}$ (2) $\left(\frac{B}{A}\right)^{\frac{1}{4}}$

(3) $\left(\frac{2A}{B}\right)^{\frac{1}{4}}$ (4) $\left(\frac{B}{2A}\right)^{\frac{1}{4}}$

3. A particle P of mass m is dropped from a height h above the ground. At the same time another particle Q of the same mass is thrown vertically upwards from the ground with a speed of $\sqrt{gh}$. If they collide head-on completely elastically, the ratio of the time taken by the particle P and Q after collision to reach the ground is

(1) $\frac{1}{1+\sqrt{2}}$ (2) $1+\sqrt{2}$

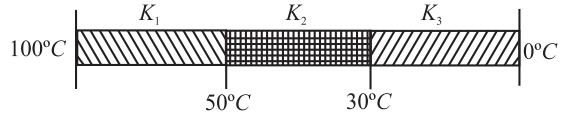
(3) $\sqrt{2}$ (4) $\frac{1}{\sqrt{2}}$

4. The mass density of a planet of radius R varies with the distance r from its centre as $\rho(r) = \rho_0\left(1 - \frac{r}{R}\right)$. Then the gravitational field is maximum at

(1) $r = \frac{R}{3}$ (2) $r = \sqrt{\frac{2}{3}}R$

(3) $r = \frac{2R}{3}$ (4) $r = \frac{3R}{2}$

5. Three rods of identical cross-section and lengths are made of three different materials of thermal conductivity K_1, K_2 and K_3, respectively. They are joined together at their ends to make a long rod (see figure). One end of the long rod is maintained at 100°C and the other at 0°C (see figure). If the joints of the rod are at 50°C and 30°C in steady state and there is no loss of energy from the surface of the rod, the correct relationship between K_1, K_2 and K_3 is



(1) $K_1 : K_2 = 2 : 5$; $K_2 : K_3 = 3 : 2$

(2) $K_1 < K_2 < K_3$

(3) $K_1 : K_2 = 5 : 2$; $K_1 : K_3 = 3 : 5$

(4) $K_1 = K_2 > K_3$

6. When a particle of mass m is attached to a vertical spring of spring constant k and released, its motion is described by $y(t) = y_0 \sin^2 2\omega t$, where '$y$' is measured from the lower end of unstretched spring. Then ω is

(1) $\sqrt{\frac{5g}{y_0}}$

(2) $\frac{1}{2}\sqrt{\frac{g}{y_0}}$

(3) $\sqrt{\frac{g}{2y_0}}$

(4) $\sqrt{\frac{g}{8y_0}}$

7. Two identical electric point dipoles have dipole moments $\vec{p}_1 = p\hat{i}$ and $\vec{p}_2 = -p\hat{i}$ are held on the x axis at distance '$2a$' from each other. When released, they move along the x-axis with the direction of their dipole moments remaining unchanged. If the mass of each dipole is 'm', their speed when they are infinitely far apart is

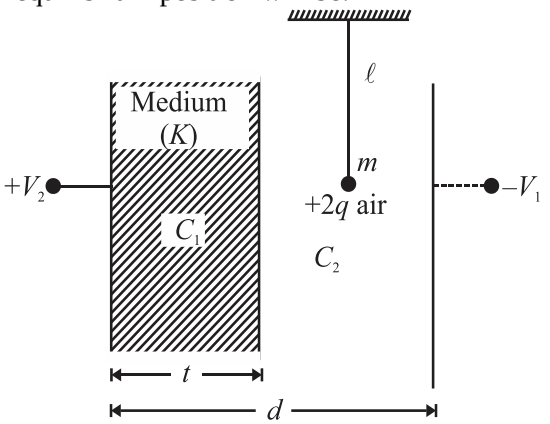
(1) $\frac{p}{a}\sqrt{\frac{1}{\pi \in_0 ma}}$

(2) $\frac{p}{4a}\sqrt{\frac{1}{\pi \in_0 ma}}$

(3) $\frac{p}{2a}\sqrt{\frac{1}{\pi \in_0 ma}}$

(4) $\frac{p}{a}\sqrt{\frac{1}{2\pi \in_0 ma}}$

8. A simple pendulum of mass 'm', length 'ℓ' and charge '$+2q$' suspended in the electric field produced by two conducting parallel plates as shown. The value of deflection of pendulum in equilibrium position will be:



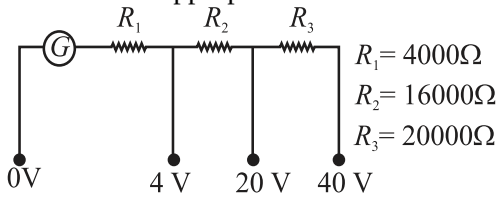
(1) $\tan^{-1}\left[\frac{2q}{mg}\times\frac{KC_2(V_2-V_1)}{(C_1+C_2)\,t}\right]$

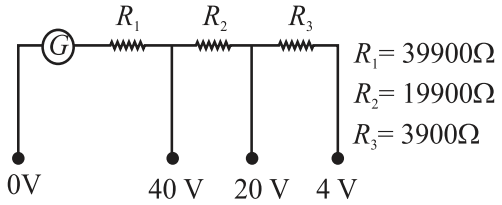
(2) $\tan^{-1}\left[\frac{2q}{mg}\times\frac{KC_1(V_1+V_2)}{(C_1+C_2)(d-t)}\right]$

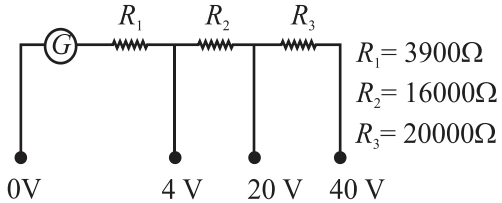
(3) $\tan^{-1}\left[\frac{2q}{mg}\times\frac{KC_1(V_2-V_1)}{(C_1+C_2)(d-t)}\right]$

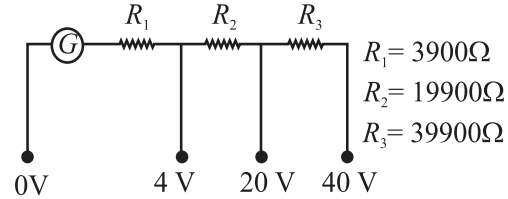
(4) $\tan^{-1}\left[\frac{2q}{mg}\times\frac{KC_2(V_1+V_2)}{(C_1+C_2)t}\right]$

9. A galvanometer of resistance 100 **Ω** has 50 divisions on its scale and has sensitivity of 20μA/ division. It is to be converted to a voltmeter with three ranges, of 0 – 4 V, 0 – 20 V and 0 – 40 V. The appropriate circuit to do so is:

(1)


(2)


(3)


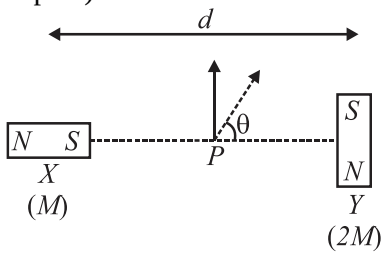
(4)


10. A particle of mass m and charge q has an initial velocity $\vec{v}=v_0\hat{j}$. If an electric field $\vec{E}=2E_0\hat{i}$ and magnetic field $\vec{B}=B_0\hat{i}$ acts on the particle, its speed will be double after a time

(1) $\frac{mv_0}{qE_0}$ (2) $\frac{3mv_0}{2qE_0}$

(3) $\frac{\sqrt{2}mv_0}{qE_0}$ (4) $\frac{\sqrt{3}mv_0}{2qE_0}$

11. Two magnetic dipoles X and Y are placed at a separation d, with their axes perpendicular to each other. The dipole moment of Y is twice that of X. A particle of charge q is passing through their mid-point P with velocity v, at angle θ = 45° with the horizontal line as shown in the figure. What would be the magnitude of force on the particle at that instant? (d is much larger than the dimensions of the dipole)



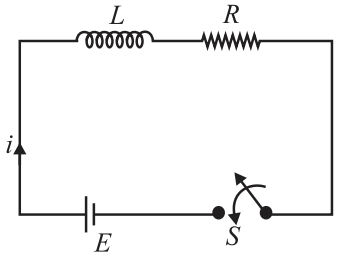
(1) $2\sqrt{2}\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3}\times qv$

(2) $\left(\frac{\mu_0}{4\pi}\right)\frac{2M}{(d/2)^3}\times qv$

(3) $\left(\frac{\mu_0}{4\pi}\right)\frac{M}{(d/2)^3}\times qv$

(4) 0

12. Consider the LR circuit shown in the figure. If the switch S is closed at t = 0 then the amount of charge that passes through the battery between t = 0 and $t=\frac{2L}{R}$ is [e = 2.7]



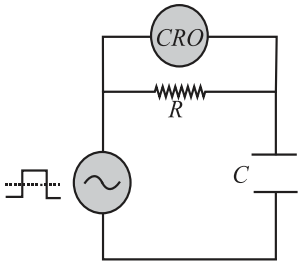
(1) $\frac{EL}{7.3R^2}$

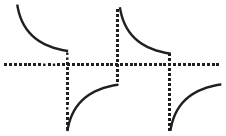
(2) $\frac{EL}{2.7R^2}$

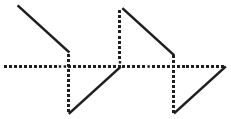
(3) $\frac{829EL}{729R^2}$

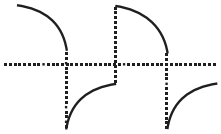
(4) $\frac{2.7EL}{R^2}$

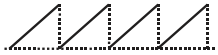
13. An RC circuit as shown in the figure is driven by an AC source generating a square wave. The output wave pattern monitored by CRO would look close to:



(1) 

(2) 

(3) 

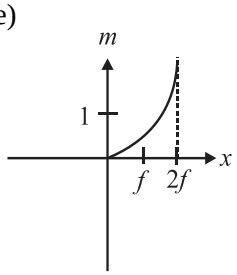
(4) 

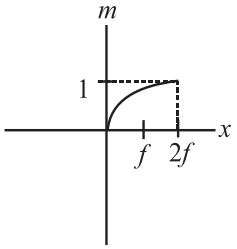
14. An electromagnetic wave is represented by the electric field $\vec{E} = -E_0\hat{n}\sin\left[\omega t + (6y - 4z)\right]$. Taking unit vectors in x, y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation $\hat{s}$ is:

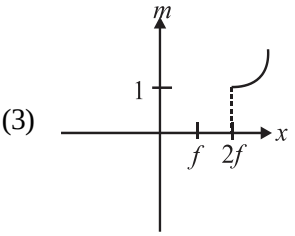
(1) $\hat{s} = \left(\dfrac{2\hat{k} - 3\hat{j}}{\sqrt{13}}\right)$ (2) $\hat{s} = \left(\dfrac{-2\hat{k} - 3\hat{j}}{\sqrt{13}}\right)$

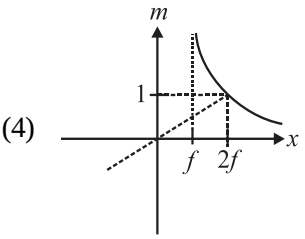
(3) $\hat{s} = \left(\dfrac{3\hat{j} - 2\hat{k}}{\sqrt{13}}\right)$ (4) $\hat{s} = \left(\dfrac{2\hat{k} + 3\hat{j}}{\sqrt{13}}\right)$

15. An object is moving from infinity towards the centre of curvature of the concave mirror along the axis of the mirror. The graphical representation of the magnitude of linear magnification (m) versus magnitude of distance of the object from the mirror (x) is correctly given by (Graphs are drawn schematically and are not to scale)

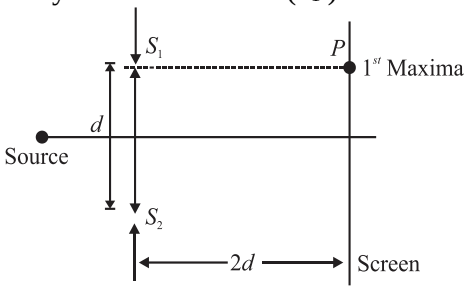
(1) 

(2) 

(3) 

(4) 

16. Consider a Young's double slit experiment as shown in figure. What should be the slit separation d in terms of wavelength λ such that the first maxima after central maxima, occurs directly in front of the slit (S_1)?



(1) $\dfrac{\lambda}{2(\sqrt{5}-2)}$ (2) $\dfrac{\lambda}{(\sqrt{5}-2)}$

(3) $\dfrac{\lambda}{2(5-\sqrt{2})}$ (4) $\dfrac{\lambda}{(5-\sqrt{2})}$

17. Particle A of mass $m_A = \dfrac{m}{2}$ moving along the x-axis with velocity v_0 collides elastically with another particle B at rest having mass $m_B = m$. If both particles move along the x-axis after the collision, the change $\Delta\lambda$ in de-Broglie wavelength of particle A, in terms of its de-Broglie wavelength (λ_0) before collision is

(1) $\Delta\lambda = 2\lambda_0$ (2) $\Delta\lambda = 4\lambda_0$

(3) $\Delta\lambda = \dfrac{3}{2}\lambda_0$ (4) $\Delta\lambda = \dfrac{5}{2}\lambda_0$

18. An ideal gas is enclosed in a cylinder made of movable walls, at pressure of 2 atm and temperature 300 K. The mean time between two successive collisions is 6×10^{-8} s. After a certain process, if the final pressure is doubled and final temperature is increased to 900 K, the mean time between two successive collisions will be close to:

(1) $3\sqrt{3}\times10^{-8}$s (2) 3×10^{-8} s

(3) $\sqrt{3}\times10^{-8}$s (4) $\dfrac{1}{\sqrt{3}}\times10^{-8}$s

19. Read the following statements:

(A) Volume of the nucleus is directly proportional to the mass number

(B) Volume of the nucleus is independent of mass number.

(C) Mass of the nucleus is directly proportional to the mass number

(D) Density of nucleus is directly proportional to the cube root of the mass number

(E) Density of the nucleus is independent of the mass number

Choose the correct option from the following options.

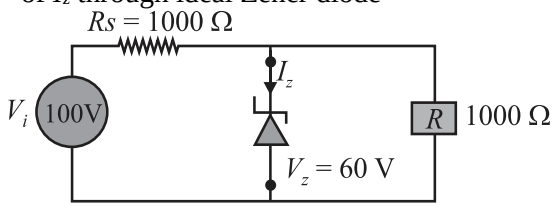
(1) (A), (C) and (D) only.

(2) (A), (C) and (E) only.

(3) (B) and (E) only.

(4) (A) and (C) only.

20. For the circuit shown below, calculate the value of I_z through ideal Zener diode



(1) 10 mA (2) 25 mA

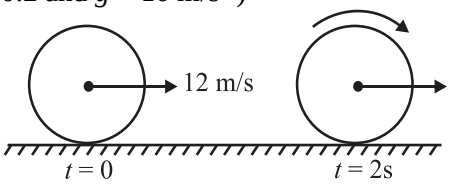
(3) 0 mA (4) 5 mA

Integer Type Questions

21. An open organ pipe of length L and a closed organ pipe contain gases of densities ρ_1 and ρ_2 respectively. The compressibility of gases are equal in both the pipes. Both the pipes are vibrating in their first overtone with same frequency. The length of the closed pipe is $\frac{x}{4}L\sqrt{\frac{\rho_1}{\rho_2}}$ where x is (Round off to the Nearest Integer)

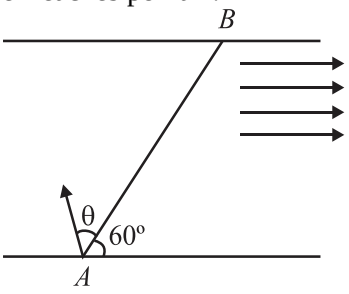
22. A tube of length 50 cm is filled completely with an incompressible liquid of mass 250 g closed both ends. The tube is then rotated in horizontal plane about one of its ends with an uniform angular velocity $x\sqrt{F}$ rad s^{-1}. If F be the force exerted by the liquid (in SI units) at the other end then the value of x will be _____ .

23. A uniform solid cylinder of mass 0.5 kg and radius r is projected with velocity 12 m/s at t = 0 s on a rough horizontal surface. It starts off with a purely sliding motion at t = 0 s. After 2 s it acquires a purely rolling motion (see figure). The total kinetic energy of the cylinder after 2 s will be _____ J (given, coefficient of friction is 0.2 and g = 10 m/s^2)



24. A circular conducting coil of radius 1 m being heated by the change of magnetic field B passing perpendicular to the plane in which the coil is laid. The resistance of the coil is 4μΩ. The magnetic field is slowly switched off such that starting at t = 0, its magnitude changes with time t (in seconds) as $B=\frac{4}{\pi}\times10^{-3}\left(1-\frac{t}{100}\right)$Tesla . The energy dissipated by the coil, before the magnetic field is switched off completely, is E = _____ mJ

25. A swimmer wants to cross a river starting from point A to land directly at point B. Line AB makes an angle of 60° with the flow of river. Magnitude of velocity of the swimmer in still water is same as that of the river. The angle θ with the line AB should be _____ degrees, so that the swimmer reaches point B.



SECTION-II (CHEMISTRY)

Single Correct Type Questions

26. Given below are two statements: One is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Paper chromatography is a type of adsorption chromatography.

Reason (R): In paper chromatography Water trapped in the paper acts as stationary phase and solvent acts as mobile phase.

In the light of the above statements, choose the correct answer from the options given below.

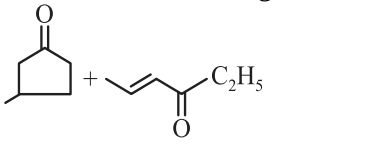
(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

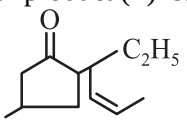
(4) A is false but R is true

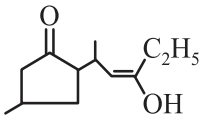
27. Consider the following reaction conditions:

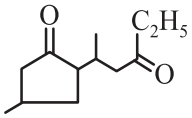


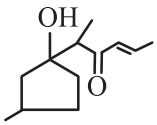
$\xrightarrow[2.H_2O/H^+]{1.EtONa}$ Major product (P)

Major product (P) is:

(1) 

(2) 

(3) 

(4) 

28. Presence of Cu^{2+} ion in a salt can be confirmed by

(a) Black PPt of CuS with H_2S/H^+

(b) Yellow PPt of $Cu(CN)_2$ with KCN

(c) Blue PPt of $Cu(OH)_2$ with NaOH

(1) a only (2) a, b only

(3) b, c only (4) a, b, c

29. The mole fraction of H_2SO_4 in an aqueous solution having conc. of 49% H_2SO_4 by mass.

(1) 0.18 (2) 8.82

(3) 0.85 (4) 0.15

30. The electrochemical cell having maximum value of standard cell potential is:

(at 298 K temperature)

Given: $E^\circ_{Zn^{2+}/Zn} = -0.76V;$

$E^\circ_{Au^{3+}/Au} = 1.40V; E^\circ_{Ag^+/Ag} = 0.80V;$

$E^\circ_{Fe^{2+}/Fe} = -0.44V$

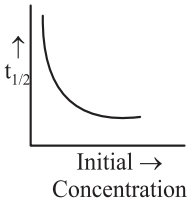
(1) $Zn(s)\left|Zn^{2+}(aq)\right|\left|Au^{3+}(aq)\right|Au(s)$

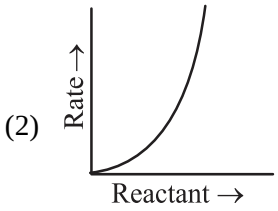
(2) $Fe(s)\left|Fe^{2+}(aq)\right|\left|Ag^{+}(aq)\right|Ag(s)$

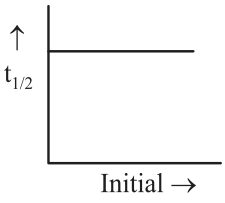
(3) $Zn(s)\left|Zn^{2+}(aq)\right|\left|Ag^{+}(aq)\right|Ag(s)$

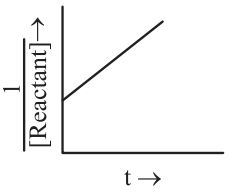
(4) $Fe(s)\left|Fe^{2+}(aq)\right|\left|Au^{3+}(aq)\right|Au(s)$

31. Which of the following graph is correct for first order kinetics :

(1) 

(2) 

(3) 

(4) 

32. The incorrect statement about diatomic oxygen is:

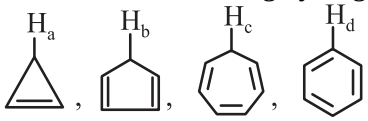
(1) It is paramagnetic

(2) The LUMO in O_2 is π^*_{2px} & π^*_{2py}

(3) It has 1 sigma and 1 Pi bond

(4) Bond order of oxygen is 2

33. Which of the following hydrogen is most acidic?

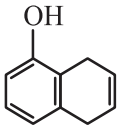


(1) H_a

(2) H_b

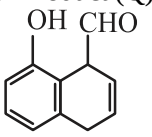
(3) H_c

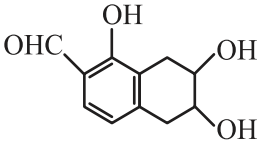
(4) H_d

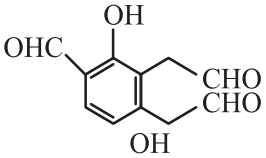
34. 

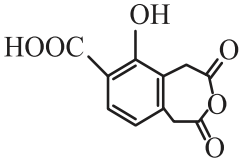
$\xrightarrow{CHCl_3+KOH}$ P (Major Product) $\xrightarrow[\text{alkaline } KMnO_4]{\text{Cold.dil.}}$ Q (Major Product).

Major Product (Q) is:

(1) 

(2) 

(3) 

(4) 

35. A Cr^{3+} ion has a spin-only magnetic moment that can be expressed as $z \times 10^{-1}$ Bohr magnetons (BM). After calculating, round off the value of z to the nearest whole number. z is

(1) 25
(2) 39
(3) 49
(4) 64

36. Match column-I with column-II.

Column-I (Isomeric pairs)		**Column-II (Type of Isomerism)**	
A	n- Butane and isobutane	P	Ring- chain isomers
B	Methoxy ethane and propanol	Q	Tautomerism
C	2- Butene and cyclobutane	R	Functional isomers
D	Ethanimine and Ethenamine	S	Chain isomers

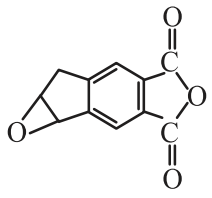
Choose the correct answer from the options given below:

	A	**B**	**C**	**D**
(1)	P	R	Q	S
(2)	R	Q	P	S
(3)	S	R	P	Q
(4)	S	R	Q	P

37. Glucose is the C-4 epimer of

(1) D- Galactose (2) D- Mannose
(3) D- Glucose (4) D-Altrose

38. When reacting the given organic compound with an excess of Grignard reagent, how many moles of the Grignard reagent are consumed per mole of the compound.



(1) 2 (2) 3
(3) 4 (4) 5

39. Which of the following is not equivalent to 2.7099×10^{24} NH_3 molecules? ($N_A = 6.022 \times 10^{23}$)

(1) 4.5 moles (2) 10.84×10^{25} atoms
(3) 76.5 g (4) 100.8 L at STP

40. The minimum wavelength of light that can produce the photoelectric effect with a photosensitive metal is:
(Given: The threshold frequency of the metal is 2.0×10^{15} s^{-1} and $c = 3.0 \times 10^8$ m/s)

(1) 1.5×10^{-7} cm
(2) 1.5×10^{-7} m
(3) 2.5×10^{-7} m
(4) 1.0×10^{-7} m

41. **Statement-I** Acidic nature of oxides increases down the group and decreases from left to right in the periodic table.
Statement-II Metallic character increases top to bottom in the periodic table.

(1) Both the statements are correct
(2) Statement I is correct but statement II is incorrect
(3) Both the statements are incorrect
(4) Statement I is incorrect but statement II is correct

42. $[Co(NH_3)_6]\ Cl_3$ has:

(1) sp^3d^2 hybridisation and high spin octahedral complex
(2) sp^3d^2 hybridisation and low spin octahedral complex
(3) d^2sp^3 hybridisation and low spin octahedral complex
(4) d^2sp^3 hybridisation and high spin octahedral complex

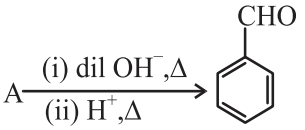
43. Consider the combustion of propane in two different systems, X and Y, as described below. What are the effects on the temperature in each system after the reaction?
System X: An isolated container that does not allow heat exchange.
System Y: A container that allows heat transfer with its surroundings.

(1) **System X:** Temperature decreases
System Y: Temperature increases
(2) **System X:** Temperature increases
System Y: Temperature remains constant
(3) **System X:** Temperature decreases
System Y: Temperature remains constant
(4) **System X:** Temperature remains constant
System Y: Temperature increases

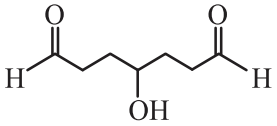
44. What is the difference in the oxidation state of sulphur in sulphate ion and sulphur dioxide?

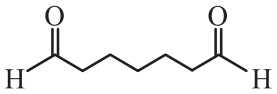
(1) 1 (2) 3
(3) 4 (4) 2

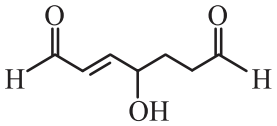
45. Consider the following reaction:

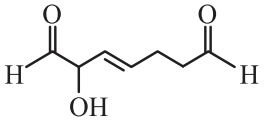


A is

(1) 

(2) 

(3) 

(4) 

Integer Type Questions

46. Consider two salts, P_2Q and R_3S. Salt P_2Q has a solubility product of 5.0×10^{-9}, while salt R_3S has a solubility product of 2.0×10^{-6}. If the solubility of P_2Q in water is S_P mol/L and the solubility of R_3S in water is S_R mol/L. If the ratio of their solubilities, $\left(\frac{S_P}{S_R}\right)$ is x then find the value of 1000x. (Nearest integer)

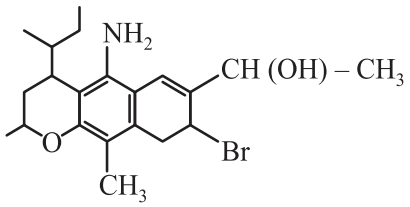
[Given: $\sqrt[3]{1.25}=1.08, \sqrt[4]{7.41}=1.65$]

47. A solution of the complex $FeCl_3.4(H_2O).(NH_3)$ shows a conductivity corresponding to 1:2 electrolyte. If the oxidation state of the central metal ion is x and the no. of chlorines coordinated directly to the central metal is/are y. Then $\frac{3x+6y}{5}$ is

48. Given : $E^{\circ}_{Cu^{2+}/Cu}=+0.34V$ and $E^{\circ}_{Cu^{+}/Cu}=+0.52V$. If ΔG° for $Cu^{2+}|Cu^{+}$ is $-x\times10^{2}$ J/mol. x is …… (answer to the nearest integer value) (1F = 96,000C) (T = 298 K)

49. You are studying the reaction between two gases, M_2 and N_2, under different temperature conditions. You observe that the reaction has a rate constant of $1.8\times10^{-4}\ dm^3\ mol^{-1}\ s^{-1}$ at a temperature of 320°C, and the rate constant increases to $0.9\ dm^3\ mol^{-1}\ s^{-1}$ when the temperature is raised to 550°C.
Given that the universal gas constant R is $8.31\ J\ K^{-1}\ mol^{-1}$. Calculate the activation energy E_a for this reaction, expressing your answer in $kJ\ mol^{-1}$. (Answer to the nearest integer value)
[Given: log (5) : 0.7, lnX = 2.3 log X]

50. The number of chiral centre(s) in the given compound is



SECTION-III (MATHEMATICS)

Single Correct Type Questions

51. Let the point $(p, p+1)$ lie inside the region $E=\left\{(x,y): 4-x\le y\le\sqrt{16-x^2}, 0\le x\le 4\right\}$. If the set of all values of p is in the interval $[a, b]$, then b^2+b-a is

(1) 3 (2) 6
(3) 2 (4) 1

52. Let $a\in R$ and α,β be the roots of the equation $x^2+70^{\frac{1}{4}}x+a=0$. If $\alpha^4+\beta^4=-30$, then the product of all possible values of a is

(1) 45 (2) 50
(3) 35 (4) None of above

53. If $\frac{1}{(20-a)(40-a)}+\frac{1}{(40-a)(60-a)}+....+\frac{1}{(380-a)(400-a)}=\frac{1}{256}$, then absolute value of the product of all values of a is:

(1) 2025 (2) 4064
(3) 3136 (4) 3025

54. The minimum number of elements that must be added to the relation $R = \{(a, b), (b, c)\}$ on the set $\{a, b, c\}$ so that it becomes equivalence is

(1) 6
(2) 7
(3) 1
(4) 3

55. The mean and standard deviation of the marks of 10 students were found to be 50 and 12 respectively. Later, it was observed that two

marks 15 and 25 were wrongly read as 40 and 50 respectively. Then the correct variance is

(1) 23
(2) 219
(3) 117
(4) 294

56. Let z be a complex number such that $\left|\frac{z-4i}{z+i}\right|=4, z\neq -i$. Then z lies on the circle with centre

(1) $(2, 0)$ (2) $\left(0,-\frac{4}{3}\right)$

(3) $\left(0, \frac{4}{3}\right)$ (4) $(0, -2)$

57. All the letters of the word NALANDA are written in all possible orders and these words are written as in a dictionary with serial numbers. Then the serial number of the word NALANDA is

(1) 462 (2) 342
(3) 354 (4) 345

58. Consider the lines L_1 and L_2 given by

$L_1: \frac{x-1}{2}=\frac{y-3}{1}=\frac{z-2}{2}$

$L_2: \frac{x-2}{1}=\frac{y-2}{2}=\frac{z-3}{3}$

A line L_3 having direction ratios $\langle 1, -2, -2\rangle$ intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(1) 6 (2) 12
(3) 18 (4) 24

59. Let $\left(a+bx+cx^2\right)^{10}=\sum_{i=0}^{20} p_i x^i, a,b,c,\in\mathbb{N}$. If $p_1 = 20$ and $p_2 = 360$, then $(a + b + c)$ is equal to

(1) 30 (2) 21
(3) 15 (4) 42

60. Let an ellipse with centre (1, 0) and latus rectum of length $\frac{1}{2}$ have its major axis along x-axis. If its minor axis subtends an angle 90º at the foci, then twice the sum of the square of the lengths of its minor and major axes is equal to

(1) 1 (2) 2
(3) 3 (4) 4

61. Let a biased coin be tossed 5 times. If the probability of getting 4 heads is equal to the probability of getting 3 heads, then the probability of getting atmost two heads is

(1) $\frac{49}{3^5}$ (2) $\frac{26}{3^5}$

(3) $\frac{102}{3^5}$ (4) $\frac{17}{3^4}$

62. Let $A=\begin{pmatrix}1 & 2\\ -2 & -5\end{pmatrix}$. Let $\alpha,\beta\in R$ be such that $\alpha A^2+\beta A=I$, Then $\alpha+\beta$ is equal to

(1) –5 (2) –3
(3) 3 (4) 5

63. The complete set of values of x for which $f(x) = \ln(x^2 – 3x + 2)$ is increasing is

(1) $\left(1,\frac{3}{2}\right)\cup(2,\infty)$ (2) $(-\infty,1)\cup\left(\frac{3}{2},2\right)$

(3) $\left(1,\frac{3}{2}\right)$ (4) $(2, \infty)$

64. The number of elements in the set $S=\left\{\theta\in\left(0,\frac{\pi}{2}\right): 3\cos^2 2\theta+6\cos 2\theta-10\cos^2\theta+5=0\right\}$ is

(1) 1 (2) 2
(3) 3 (4) 4

65. For $k\in R$, let the solutions of the equation $\cos(\sin^{-1}(x\cot(\tan^{-1}(\cos(\sin^{-1}x)))))=k$, $0<|x|<\frac{1}{\sqrt{2}}$ be α and β, where the inverse trigonometric functions take only principal values. If the solution of the equation $x^2-bx-5=0$ are $\frac{1}{\alpha^2}+\frac{1}{\beta^2}$ and $\frac{\alpha}{\beta}$, then b is equal to

(1) 6 (2) 12
(3) 4 (4) $\frac{1}{6}$

66. If domain of the function $\log_e\left(\frac{6x^2+5x+1}{2x-1}\right)+\cos^{-1}\left(\frac{2x^2-3x+4}{3x-5}\right)$ is $(\alpha,\beta)\cup(\gamma,\delta]$, then $27\left(\frac{\alpha^2+\beta^2}{\gamma^2+\delta^2}\right)$ is equal to

(1) 10 (2) 20
(3) 13 (4) 17

67. Let $\alpha = \lim\limits_{x\to\frac{\pi}{4}} \frac{\tan^3 x - \tan x}{\cos\left(x+\frac{\pi}{4}\right)}$ and $\beta = \lim\limits_{x\to 0}(\cos x)^{\cot x}$. The quadratic equation with α, β as its roots is

(1) $x^2+3x-4=0$ (2) $x^2-3x+4=0$

(3) $x^2+4x+3=0$ (4) $x^2-4x+3=0$

68. The number of points, where the function $f: R \to R$, $f(x) = |x-1|\cos|x-2|\sin|x-1| + |x-3|\,|x^2-5x+4|$, is not differentiable is

(1) 1 (2) 2

(3) 3 (4) 4

69. In an office 70% people like tea and 60% likes coffee then percentage of people who like both tea and coffee is (all people like to drink tea or coffee)

(1) 30 (2) 40

(3) 60 (4) 80

70. **S-I:** $\int_0^{2\pi} \sin x\,dx = 0$

S-II: Area bounded by $y = \sin x$, x-axis and from $x = 0$ to π is zero

(1) Only S-I is true

(2) Only S-II is true

(3) Both S-I and S-II are true

(4) Both S-I and S-II are false

Integer Type Questions

71. If the area of the bounded region $R = \left\{(x,y): \max\{0, \log_e x\} \le y \le 3^x, \frac{1}{3} \le x \le 3\right\}$ is $\alpha(\log_e 3)^{-1} + \beta(\log_e 3) + \gamma$, then the value of $\left(\alpha + 2\beta - 4\gamma + \sqrt[3]{3}\right)^2$ is equal to

72. Let $y = y(x)$ be a solution curve of the differential equation, $2y^3dx + 2x\left(x^2-y^2\right)dy = 0$ such that $y(1) = 1$, then $\frac{\left(y(2)\right)^2 - 8\ln y(2)}{2}$ is equal to

73. Let the points A and B be respectively the circumcentre and the orthocentre of a ΔPQR. If $\overrightarrow{AP} + \overrightarrow{AQ} + \overrightarrow{AR} = \lambda\overrightarrow{AB}$, then λ is

74. Let the centre of a circle C be (α,β) and its radius $r < 8$ where $\alpha, \beta \in Z$. Let the lines $3x + 4y = 24$ and $3x - 4y = 32$ touches circle and $4x + 3y = 1$ be a line passing through centre of circle. Then radius of circle is equal to

75. The remainder when $(49)^{1011} + (27)^{674}$ is divided by 5 is:

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